

# TASK 2 – NETWORKING & COMPUTE

## *Terraform-Managed Azure Subscription Baseline*

### 1. Objective

To design and provision the core Azure networking and compute infrastructure using Terraform. This task covers Week 3 and Week 4 and establishes the network path required for deploying a Linux virtual machine.

### 2. Week 3 – Networking Infrastructure

The networking layer consists of an Azure Virtual Network with address space **10.0.0.0/16**, a subnet named **subnet-default** using **10.0.1.0/24**, and a Network Security Group named **nsg-terraform-baseline**. The NSG contains inbound rules for HTTP on port 80 and SSH on port 22 and is associated with the subnet.

### 3. Week 4 – Compute Infrastructure

The compute layer contains a static Standard Public IP, a Network Interface attached to the subnet, and a Linux virtual machine named **vm-terraform-baseline**. The VM uses Ubuntu 22.04 LTS Gen2, the **Standard\_D2s\_v4** size, and SSH public-key authentication. The VM is deployed in Southeast Asia because the Azure for Students subscription has regional deployment restrictions for the networking and compute resources.

### 4. Network-to-Compute Flow

| Layer        | Resource                | Purpose                                    |
|--------------|-------------------------|--------------------------------------------|
| Network      | vnet-terraform-baseline | Provides the virtual network address space |
| Network      | subnet-default          | Provides the subnet for workload resources |
| Security     | nsg-terraform-baseline  | Controls inbound network traffic           |
| Connectivity | pip-terraform-baseline  | Provides a static public IP                |
| Connectivity | nic-terraform-baseline  | Connects the VM to the subnet              |
| Compute      | vm-terraform-baseline   | Runs the Linux workload                    |

### 5. Security Rules

The Network Security Group includes an inbound HTTP rule for TCP port 80 and an inbound SSH rule for TCP port 22. These rules demonstrate basic network access control for the deployed workload. The SSH source is broad in the current lab configuration and can be restricted further for a production environment.

### 6. Terraform Implementation

The infrastructure is defined as Terraform resources and later consumed through the reusable network and compute modules. Module outputs expose values such as the subnet ID and VM public IP to the rest of the root configuration.

### 7. Expected Result

A functioning Azure network and Linux compute environment is provisioned through Terraform. The VM is connected to the configured subnet through its network interface, protected by the NSG rules, and reachable through its assigned public IP.

### 8. Evidence

The accompanying README contains the recommended screenshot checklist for Task 2. Screenshots should demonstrate the VNet, subnet, NSG rules, NSG association, public IP, NIC, VM and Terraform outputs. Do not include private keys or credentials.